

# AI for Bharat Hackathon

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Team Name : therahul

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Problem Statement : AI for Retail, Commerce & Market Intelligence ( Bid-Shield AI-Powered Tender Compliance Platform)

## **Brief about the Idea:**

India's public procurement market represents USD 500-600 billion annually (15-20% of GDP), with over 75,000 government tenders released monthly. However, SMEs face a 90% failure rate in tender bidding, capturing only 25% of central procurement by value despite constituting 99% of enterprises.

The core problem: SMEs lose significant money through EMD (Earnest Money Deposit) forfeiture (1-5% of tender value), waste time on unwinnable tenders, and face technical disqualifications due to hidden "killer clauses" that cause 20-30% of rejections.

Bid-Shield uses AI to analyze tender documents, identify compliance requirements, detect potential disqualifications, and generate strategic pre-bid queries to challenge restrictive specifications that violate Central Vigilance Commission (CVC) guidelines.

## Your solution should be able to explain the following:

- How different is it from any of the other existing ideas?
- How will it be able to solve the problem?
- USP of the proposed solution

### Differentiation from Existing Solutions:

**Proactive Risk Detection:** Unlike basic tender portals that just list opportunities, Bid-Shield actively identifies "killer clauses" and hidden disqualification risks before you bid

**AI-Powered Legal Analysis:** Uses specialized AI agents (Claude Sonnet 4.5) trained on CVC guidelines, court precedents, and procurement regulations to identify violations

**Automated Pre-Bid Query Generation:** Generates legally sound challenges to restrictive specifications with proper citations and legal precedents

**Financial Impact Assessment:** Calculates EMD exposure, success probability, and ROI for each tender opportunity

**MSME-Specific Benefits:** Automatically identifies exemptions and benefits available to registered MSMEs

### How It Solves the Problem:

**Reduces EMD Losses:** Identifies unwinnable tenders early, preventing costly EMD forfeiture

**Improves Win Rate:** Target improvement from industry average <10% to 20%+ through better tender selection

**Saves Time:** Automates 50+ hours of manual analysis per tender through AI-powered document processing

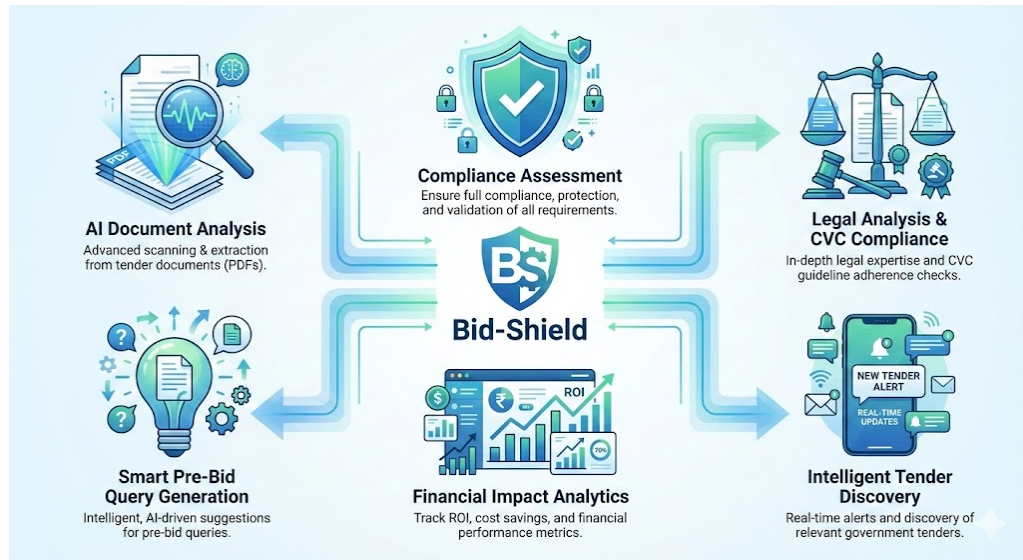
**Levels the Playing Field:** Empowers SMEs to challenge unfair specifications that favor large enterprises

### USP (Unique Selling Proposition):

"The only AI platform that not only tells you IF you can win a tender, but also HOW to challenge unfair specifications to improve your chances - backed by legal precedents and CVC guidelines."

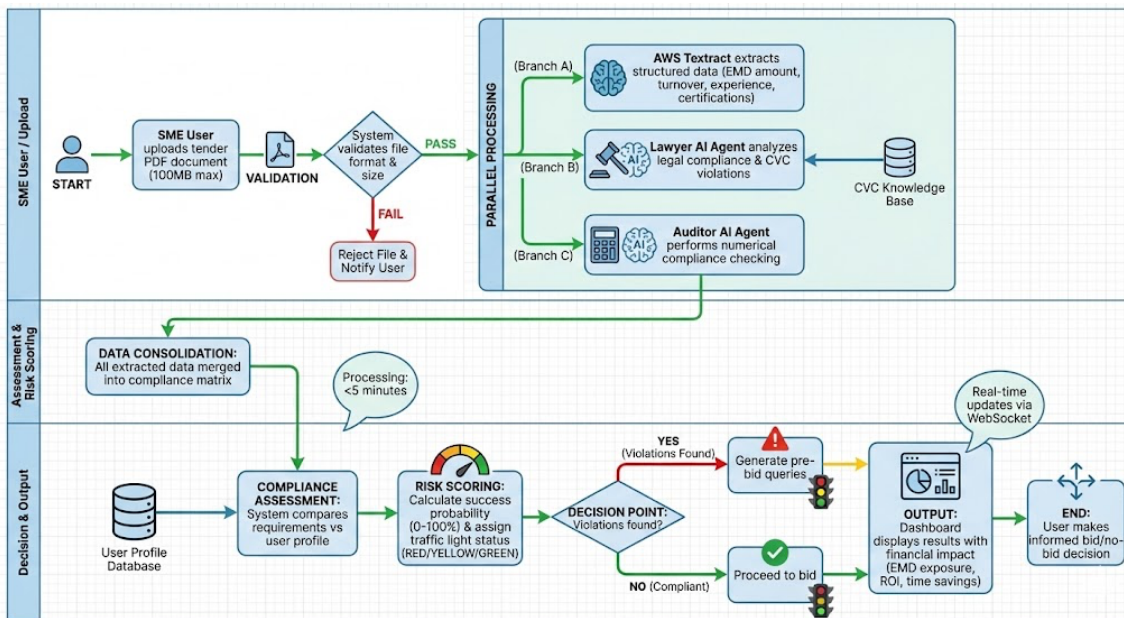
## List of features offered by the solution

It is always better to add a few visual representations (drawings/sketches/illustrations etc.) to your presentation, it adds to the power through which it reaches the audience.



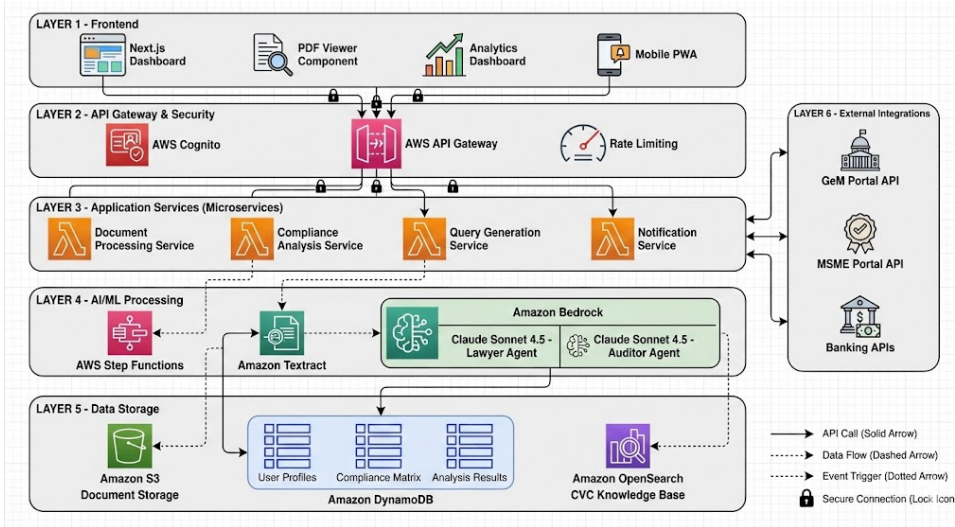
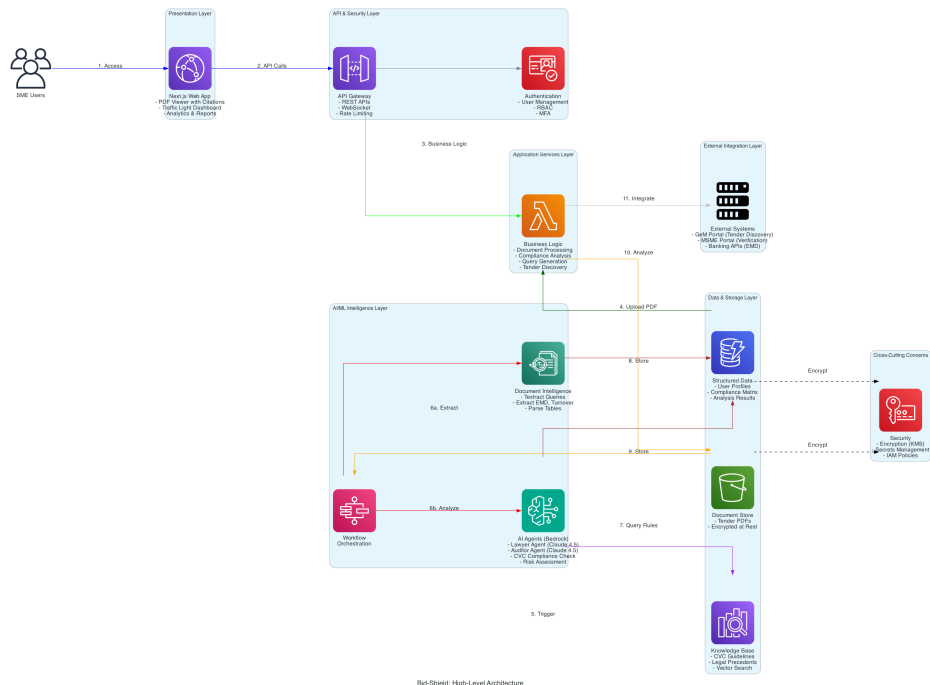
## Process flow diagram or Use-case diagram

Add a flow diagram or a use case diagram or an architecture diagram.



## **Wireframes/Mock diagrams of the proposed solution (optional)**

## Architecture diagram of the proposed solution:



## Technologies to be used in the solution:

### Frontend Technologies:

Next.js 16 (React framework with server-side rendering)

TypeScript (type-safe development)

Tailwind CSS (responsive UI design)

React PDF Viewer (document visualization)

Chart.js / Recharts (analytics visualization)

WebSocket API (real-time updates)

### Backend Technologies:

AWS Lambda (serverless compute)

AWS API Gateway (RESTful API management)

AWS Step Functions (workflow orchestration)

Node.js / Python (Lambda runtime environments)

### AI/ML Services:

Amazon Textract (document text extraction with Queries feature)

Amazon Bedrock (Claude Sonnet 4.5 for AI agents)

Amazon OpenSearch (vector database for CVC knowledge base)

### Data Storage:

Amazon S3 (document storage)

Amazon DynamoDB (NoSQL database for user profiles and analysis results)

Amazon ElastiCache (Redis for caching)

### Security & Authentication:

AWS Cognito (user authentication and authorization)

AWS KMS (encryption key management)

AWS WAF (web application firewall)

### DevOps & Monitoring:

AWS CloudFormation / CDK (infrastructure as code)

AWS CloudWatch (monitoring and logging)

AWS X-Ray (distributed tracing)

GitHub Actions (CI/CD pipeline)



## Estimated implementation cost (optional):

**Add as per the requirements for the hackathon:**

Innovation partner **I12S**  
HACKATHON

Media partner **YOURSTORY**

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Thank You

